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Innovative Workflow for Daily Injection Gas Breakthrough Quantification: Integrating Reservoir Simulation Molecular Source Tracking with Surface Separation Simulation While Utilizing the Power of Virtual Flow Models

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Abstract

This paper introduces a novel method to quantify the sources of surface products (oil & gas) by integrating sub-surface simulation with regular well tests. It uses molar compositions and production contributions from each well at the end of history-match model results, assimilating them into a source-tagged equation of state (EOS) model for surface separation simulation. This approach tracks each component, and its sources, to the final products, allowing accurate quantification of each source's contribution. Specifically, it determines the amount of injection gas breaking through and producing with each Mscf of gas and quantifies the condensate produced with oil by tracking the moles originating from the reservoir gas cap.

Modern reservoir simulation software offers tracking options to quantify the molar fraction of each component from each source (reservoir oil, reservoir gas cap, and injection gas) for each producing well. The method aggregates these data into an "extended" composition using a tagged version of the original EOS model. As an example, the original 8-component EOS model is converted into a 24-component model, and the producing streams of each well are similarly converted. This extended composition becomes the reference composition for EOS modelling of surface well test simulations. Each component and its sources are tracked through the surface process stages, resulting in molar streams, with origin information, intact in both surface oil and gas products. The relative ratios of each source in these product streams quantify the volumes from each source, such as Injection Gas Breakthrough volumes in the produced gas and condensate in the produced oil utilizing Virtual Flow Meter (VFM) based Daily Well Rates estimation.

The computed fractions are applied to "path-to-surface" corrected standard rates of oil and gas, yielding source-specific rates during well tests. The same surface product fractions are applied to allocated daily rates when the well is not on test. This implementation of source tracking demonstrates the application of these

surface source fractions to daily averaged well rates using real-time wellhead data. This is unprecedented in two ways:

1. Daily oil and gas rates are available automatically at the end of each day and aggregated to field level rates for reconciliation against daily Actual rates.
2. Daily VFM (Virtual Flow Meter) based quantification of Injection Gas Breakthrough and Gas Cap Condensate.

This paper documents and illustrates a rigorous, yet flexible, coupling between sub-surface simulation and surface separation simulation using an innovative extended EOS model. This approach enhances the resolution and tracking of each molecule in the produced stream as it passes through the path-to-surface separation facility. The paper also describes the deployment of this novel approach in an existing online workflow used in the daily operations of eight ADNOC Onshore fields. This is likely to replace the legacy system that lacked a solid technical foundation.

Introduction

eGOR Source Tracking brings together two industry-accepted petroleum engineering applications to provide an enhanced value outcome in terms of identifying and quantifying the originating sources of surface products. Reservoir simulation is the industry standard in simulating the sub-surface mechanisms that govern the movement and production of fluids in the reservoir. It models the rock and fluid properties to a great level of detail, breaking the petroleum reservoirs into millions of grid cells and initializing them with appropriate properties and pressures. As the fluids are produced from the drilled wells according to the development plans, pressure profile in the reservoir change, which governs the movement and mixing of fluids. The reservoir can be comprised of oil phase at reservoir conditions and often accompanied by a gas phase (gas cap). Often, Gas Injection is utilized by the development strategy to maintain the pressure and to mix with the oil to promote mobility. Hence up to three phases of fluids can contribute to the produced wellstreams. Typically, when the reservoir simulation model used a compositional fluid model, the produced wellstream is in terms of moles/day of each component comprising the fluid (EOS) model.

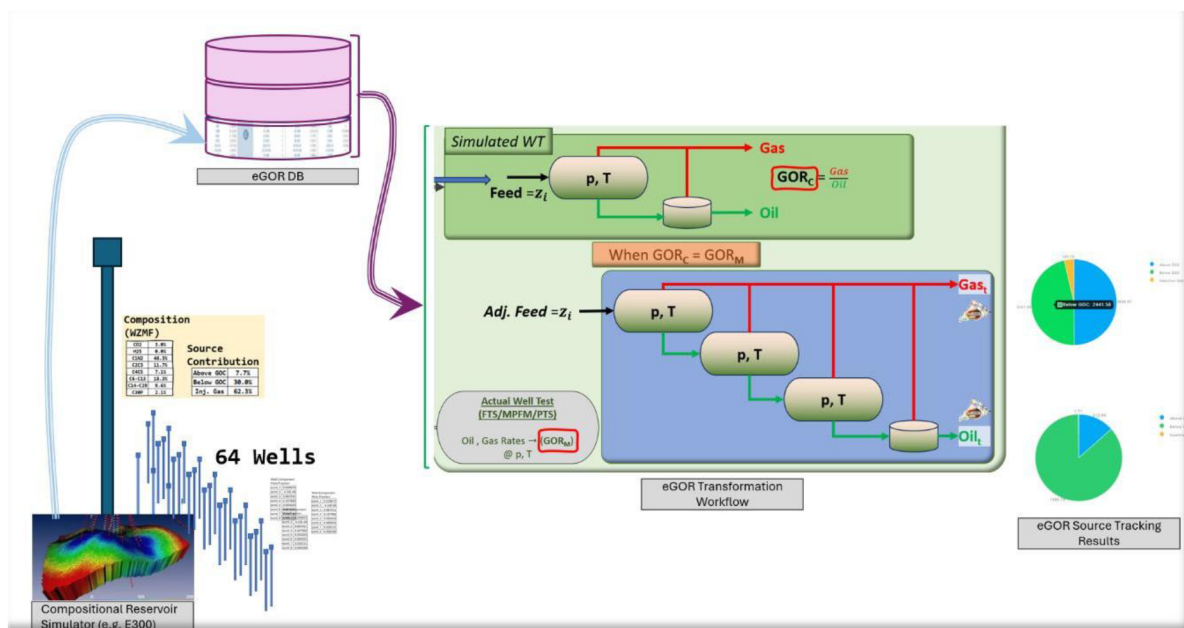


Figure 1—Depicts the high-level workflow implementation of eGOR Source Tracking

When these wellstreams are produced at the wellhead, they are typically commingled and jointly produced into Field Separators to separate the surface products of oil, gas and water. To quantify the production of individual wells, single wells are routinely (typically monthly) routed to smaller separators, or multiphase meters, to estimate well rates. eGOR is a state-of-the-art technology to transform these metered well rates to consistent standard condition rates via a relevant path-to-surface separation process. The technology employed by eGOR is surface separation process simulation using equation of state (EOS) modelling.

eGOR Source Tracking ties two industry standard technologies into a seamless solution by:

1. Using the latest well composition from a history matched simulation model as the representative reference composition.
2. Using the latest source contribution for each mole fraction from the physical tracer verified history matched model.
3. Combining the two to compute a tagged EOS model with a refined molar fraction.
4. Use standard eGOR surface process simulation workflow with the refined EOS model, tracking the tagged-molar amounts (lbmol/day) throughout the system.
5. Finally segregating the source tagged molar amount to compute respective segregated volumetric rates.

The standard eGOR workflow includes adjusting the reference composition, in step 1. to match the current well test gas-oil ratio and the gas & oil metered rates. i.e. the history matched composition is updated to match current flowing conditions. The Source-Tracking workflow performs the same surface process simulation with the tagged-EOS. Hence two crucial outcomes are achieved by the integrated modelling approach:

1. Reservoir Simulation provides the latest by-well composition with the latest source (origin) information.
2. Surface Process Simulation retains this refined molar tracking to compute field specific surface products, while quantifying the contribution from original sources.

The update to the eGOR database with this information is performed occasionally, typically once or twice, as and when the history-matched simulation is updated.

The utilization of the eGOR database information is done daily for each well test that is conducted; and then processed by eGOR.

eGOR Standard Workflow

eGOR is ADNOC Onshore's online (web-based) system to manage well testing data, its transformation from measured quantities to standard condition quantities following an integrated workflow involving three stakeholders:

1. **Operations** (OPS) review the real time data capture by the FDIS (Aveva PI) specific to the well test under review, confirming the correctness of data and performing a QC from an operational perspective
2. **Reservoir Management** (RM) team reviews the QCed data and verifies the measured quantities from fluid perspective, including the PVT files used and reservoir engineering aspects.
3. Finally, **Production Optimization** (PO) utilize the inputs from OPS & RM, coupled with well modeling and historical rates, to finally decide on the well test quality, approving or rejecting the data, and pushing the data to the corporate database for allocation and other usage.

eGOR performs EOS based transformation ensuring the applicable path-to-surface, based on which standard conditions oil & gas volumes (rates) are computed.

eGOR has been deployed in nine fields of ADNOC Onshore since early 2024 and fully leverages the real time data. The involvement of all stakeholders and the ability to trim the real time data ensures better inputs and the EOS modelling ensures representative PVT and path-to-surface transformation, ultimately ensuring more representative std. volumes (rates) for allocation purposes, contributing to reduction of reconciliation factors.

Fig. - 2 shows the various screens from this workflow.

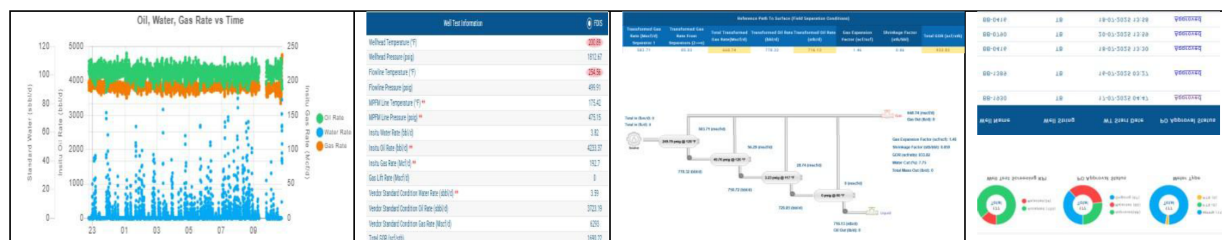


Figure 2—Screen from eGOR workflow accessed by each stakeholder to jointly process the well test, culminating an approval or a rejection.

eGOR Enhanced (Source-Tracking) Workflow

For any well test that is handled and processed by eGOR, the transformed std. oil & gas rates are computed and reported. For cases where the gas oil ratio (GOR) is in line with the solution GOR (R_{si}) of the original fluids these are the final outcomes. However, if the GOR goes well above these initial values, it indicates production of free gas (additional gas over and above the expected solution gas from the oil). Source of such free gas becomes a crucial question, and its quantification becomes a big technical challenge, with fiscal implications.

How can the source of produced volumes of oil and gas be tracked? One way is to add physical tracers into the injection gas and then capture the quantities in the producing wells. This can be done only for a handful of wells and for short duration of time. Plus, this can only identify and quantify injection gas source. Reservoir simulation utilizing an Equation of State (EOS) model provides the ability to quantify and track the movement of individual components of the EOS model as they move from the original locations and sources to the producing wells. One classical way that has been utilized in the past (Adli et. al.) is to initialize the original simulation with tagged components. The reservoir is typically split into an oil rim and a gas cap. These regions are initialized with appropriate compositions (z_i) corresponding to these phases. It is usually done with compositional gradient calculation with a clear gas oil contact (GOC). To track the movement of these original fluids, the EOS model is duplicated with tags (prefix) denoting the origin (or source). Additionally, if the reservoir is undergoing gas injection, the injection gas stream (as additional source) can be added to such a formulation to end up with 3 times the number of components tagged with appropriate sources. An example could be illustrated below:

Table 1—Component Names of Extended EOS Model used in eGOR Source Tracking

Original EOS in Sim. Model	Above GOC	Below GOC	Injection Gas
CO2	a-CO2	b-CO2	i-CO2
H2S	a-H2S	b-H2S	i-H2S
C1N2	a-C1N2	b-C1N2	i-C1N2
C2C3	a-C2C3	b-C2C3	i-C2C3
C4C5	a-C4C5	b-C4C5	i-C4C5
C6-C13	a-C6-C13	b-C6-C13	i-C6-C13
C14-C29	a-C14-C29	b-C14-C29	i-C14-C29
C30P	a-C30P	b-C30P	i-C30P

The Properties of each "incarnation" of the component remains the same ensuring the behavior of the fluid is unaltered. But the unique component derived from the original component name and a "prefix" ensures the simulation tracks and reports the molar quantities separately in each wellstream for each time step. These can be utilized in the Surface Separation EOS modelling to estimate the molar fractions of the surface products from each source, thus estimate the contribution of each source to the total volumetric rates.

This approach, however, increases the number of components three-fold and will result in a substantial increase in the runtime of the model, which needs to be re-run for the entire history match period to obtain the mole fractions at end of history.

An alternative to the explicit tripling the number of components is to use the **tracking** option offered by reservoir simulators. This essentially allows the simulation to be run with the original number of components (e.g. 8) while the simulator tracks and traces the component originating from each of the 3 sources, above GC, below GOC and Injection Gas. Simulators have in-built mechanism for tracking and reporting these vectors for above and below GOC but additional tracers need to be defined and reported for the injection gas source.

What is the additional information that is available with the Track option? This option, with its associated tracer specifications for injection stream source allows each component mole fraction to be split into the contribution from each source.

Original Composition		Source Tracking Reference Composition					
Original Composition	Mole %	Above GOC	Mole %	Below GOC	Mole %	Injection Gas	Mole %
CO2	2.8121	A-CO2	0	B-CO2	1.3553	I-CO2	1.4569
H2S	0.0005	A-H2S	0	B-H2S	0.0005	I-H2S	0
C1N2	61.2959	A-C1N2	0	B-C1N2	23.8514	I-C1N2	37.4445
C2C3	11.1419	A-C2C3	0	B-C2C3	6.2826	I-C2C3	4.8593
C4C5	6.6441	A-C4C5	0	B-C4C5	4.1919	I-C4C5	2.4522
C6-C13	11.9166	A-C6-C13	0	B-C6-C13	9.926	I-C6-C13	1.9906
C14-C29	5.0088	A-C14-C29	0	B-C14-C29	5.0088	I-C14-C29	0
C30P	1.1575	A-C30P	0	B-C30P	1.1575	I-C30P	0

Figure 3—Original EOS split into 3 x the no. of components in eGOR.

This usage is controlled by the user selection of enabling Source Tracking in eGOR. Once this is done, the 8 component mole fractions are converted into a 24-component mole fraction and the surface simulation, including the Target-GOR matching and Multi-Stage Separation Process uses this tagged EOS model. The molar information including the source information is retained throughout the surface simulation at each stage and the incoming stream is split into the gas and oil phases. Similarly, the source tagged moles are aggregated for the gas liberated from each stage of separation. As illustrated in Fig. - 4 the total gas and

the net oil molar streams still have the source information available and can be segregated before being converted to volumes at std. conditions.

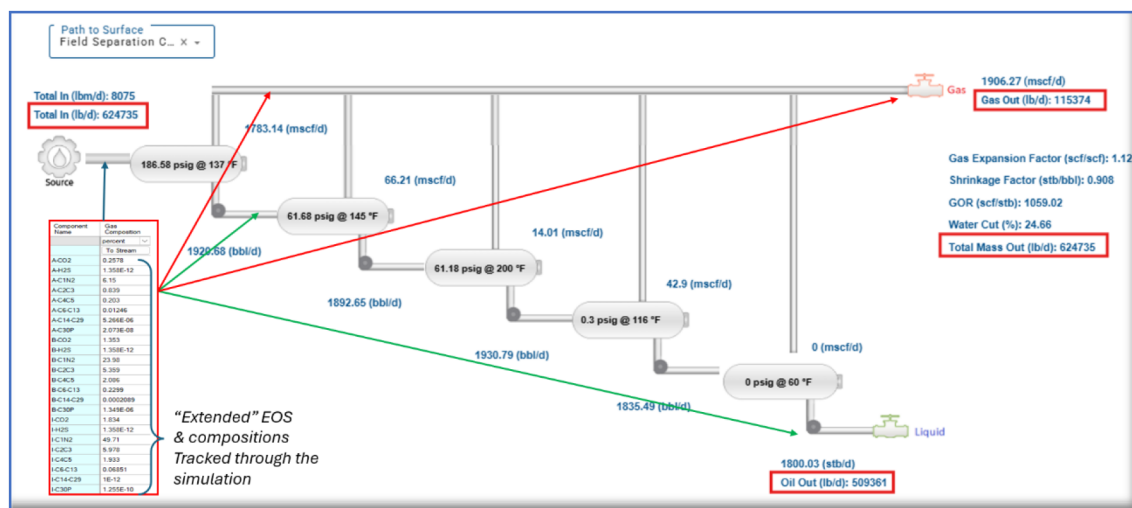


Figure 4—The multi-stage separation of the standard eGOR workflow ensures correct path-to-surface and corresponding std. volume. For Source Tracking workflow, the simulation is performed with the "extended" (e.g. 24-component) EOS model and additional mass balance is honored and displayed

On the results page, eGOR interface provides a summary of these segregated volumes, and associated properties while additionally pushing these results to the database:

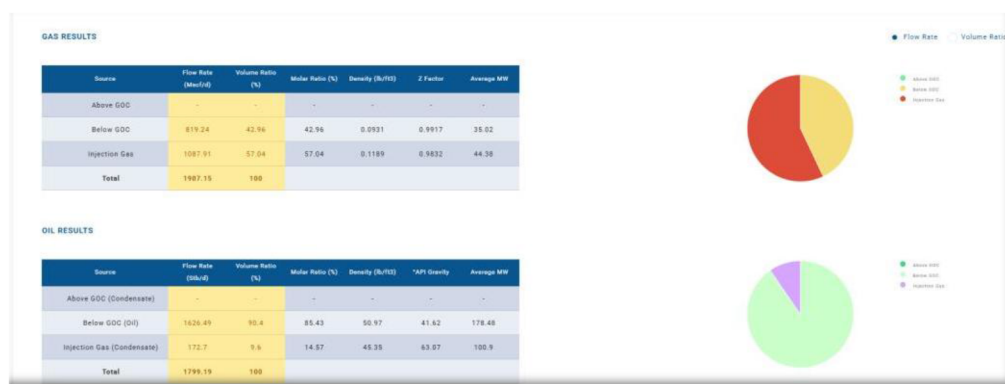


Figure 5—Result page of eGOR Source Tracking workflow

Reservoir Simulation

This section documents the features and corresponding keywords necessary to be specified in the reservoir simulation data-set for the simulation to compute and report appropriate results that are then uploaded by eGOR for use in the source tracking workflow.

These features and keywords will be documented in the context of the compositional reservoir simulator E300 from SLB, that is the most widely used simulator in the industry and in ADNOC (including ADNOC Onshore). Similar keywords and features are available for other simulators. In fact, the other popular simulator tNavigator can utilize an E300 mode in which the identical keywords are used.

At the outset, it should be clear that only compositional simulators are applicable since both sub-surface and surface modeling utilize the EOS fluid model.

TRACK

Tracking is a feature or option in compositional reservoir simulators that enables the division of hydrocarbons originally in place to be tracked as the simulation progresses and material is produced via wells. Traditionally, this is used for reporting recovered surface volumes, but in this source tracking we utilize the molar stream and not the surface volumes, since the surface volumes from the well test will be used instead.

Fig. - 6 is a screenshot from the E300 Reference Manual for this keyword:

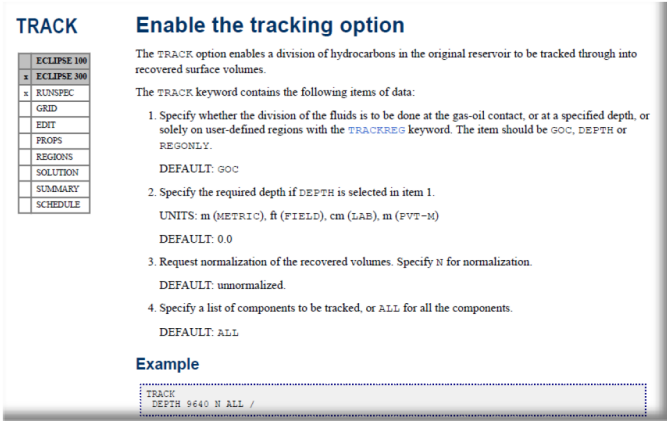


Figure 6—Documentation of TRACK option for E300. Also applicable to tNavigator running in E300 mode

We reproduce here the actual keyword usage and describe the associated data items (options) used:

TRACK		
GOC	2*	ALL

For eGOR Source Tracking, we are interested in the contribution from Reservoir Oil (below the GOC) and from the Gas Cap (above the GOC), Hence the first option to be specified to the TRACK keyword is GOC.

The second option (DEPTH) is not relevant hence we default on it.

The third option (Normalization) is important and needs discussion. However, the default is no Normalization hence the above Data files snippet achieves this using 2*, which means default the two options (Depth & Normalization).

This is convenient for normal simulator reporting purposes. However, it achieves this by normalizing similarly the source componen t molar rate vectors: WTQRA_ *n*, and WTQRB_ *n* when *n* is the component number in the EOS model. And the Vector WTQR is defined as the "Traced Molar Rate".

The final option specified could also have been defaulted: the component of the EOS model to be tracked. The default is "ALL" which has been explicitly specified. If few components were to be tracked, the names of those components would have been specified separated by spaces.

To comply with the requirements of eGOR source tracking these are the mandatory specifications with TRACK keyword:

- Item 1: GOC
- Item 2: Default
- Item 3: N
- Item 4: ALL

INJECTION GAS

If gas injection is active in the reservoir simulation, this becomes an additional source of streams in the production wells. Injection Gas is an external stream and is not tracked as part of the Above/Below GOC with A/B suffix to the traced molar rates (as in WTQRA). Hence an additional mechanism, using Tracers, is used to track the movement of injection gas components through the model and contribute to the production from wells.

In such cases, a multi-step mechanism is used:

1. Set Injector Well stream mole fractions (this is necessary if gas injection is active, irrespective of TRACK option)

```
WELLSTRE
--      CO2      H2S      C1N2      C2C3      C4C5      C6-C13      C14-C29      C30P
'GAS'    0.03     0.0     0.78     0.10     0.05     0.04      0.0       0.0      /
/
```

2. Associate the injection stream in use with the injection well or group of injection wells (using well lists as in the example below). This too is part of the based model:

```
WINJGAS
' *GIWEL'  STREAM HCGAS /
/
```

3. Define a tracer name of each component of the injection gas (i.e. each component of the EOS Model). This is additional keyword need for tracking. Tracers can be named by user e.g below.

```
TRACER
T1  C1N2 /
/
```

4. Associate each tracer name (T1 in this example) to each gas injection well or gas injection well lists, if used, using the WTRACER keyword. This too is needed for Tracking option.

```
WTRACER
--
' *GIWEL'  T1      1      /
/
```

Note that a well can be a gas injector or water-alternate-gas (WAG) injector. If it ever experiences gas injection, the tracer needs to be associated with it using a keyword like above. Make sure no gas injection wells are missed.

Well Tracer Production Concentration is specified in the SUMMARY section for each component Below example shows the tracer production concentration for 1 component for Injection Gas:

```
WTPCT1
```

KEY SUMMARY VECTORS

In E300 and similar reservoir simulators, summary vectors are keywords requested to be written to Summary files after each timestep of the simulation. These are written as a function of date or time and are the key mechanism for visualizing simulator profiles of rates, pressure or other variables of interest.

For use in eGOR Source Tracking, the following key summary vectors are mandatory:

Summary Vector	Explanation
WOPR	Well Oil Production Rate
WZMF_ <i>n</i>	Well Total Mole Fraction of component <i>n</i>
WTPCA_ <i>n</i>	Well Tracer Production Concentration from Above GOC
WTPCB_ <i>n</i>	Well Tracer Production Concentration from Below GOC
WTPCT <i>n</i>	Well Tracer Production Concentration Injection Gas source

VISUALIZATION

It is useful to view and visualize the range of various key summary vectors and their relationships with each other.

Fig. - 7 shows the range of WZMF_3 for all wells for all simulation timesteps. This corresponds to C1N2 in the example simulation EOS model. It shows that wells start at two clustered compositions of about 50% and 60% and produce at those levels for about 7 years before diverging. At end of history (31-Dec-2023) well are producing at extremely diverse compositions of C1N2 from under 45% to over 75%. This illustrates the immediate value of using E300 integration as a basis of reference composition in eGOR. The original reference compositions were based on reservoir defaults, whereas this dynamic reference composition is well specific and more representative.

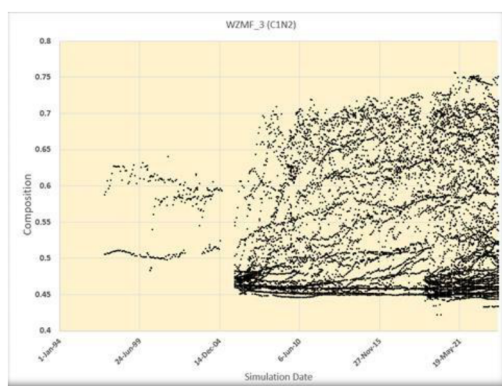


Figure 7—Range of WZMF_3 (C1N2) for all the wells for all of history

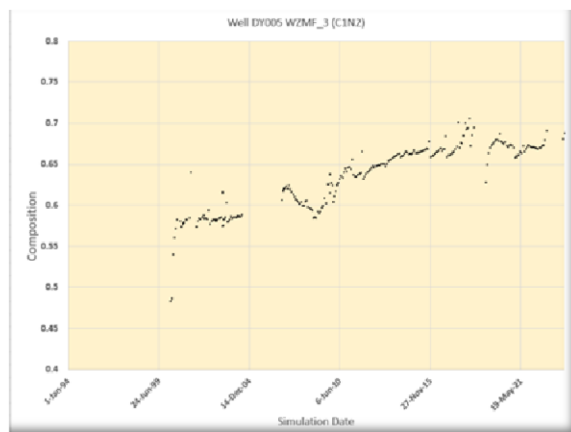


Figure 8—wzmf_3 for a single well and its development in time.

Looking at a single well, the composition for C1N2 started at about 48% in 2000 and the current level (31-Dec-2023) has reached 69%.

We now look at the Oil Production Rates Profile. Fig. - 9 shows that values for all wells in the model for all times. This extends to about 5000 STB/D. Superimposed on this plot is the rates for the same example well shown in red. This seems to be an average well with rates ranging between 1000 – 2000 STB/D.

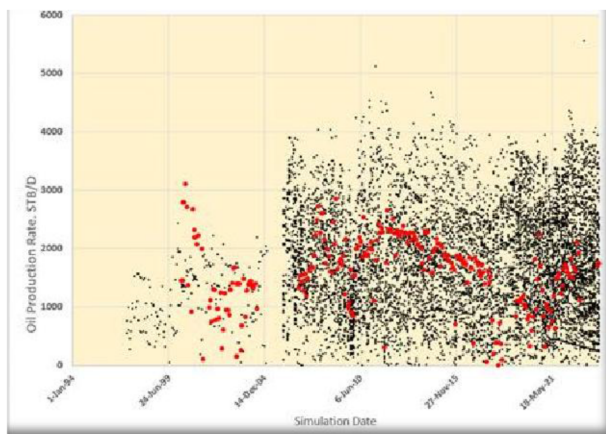


Figure 9—Oil Production rates for all wells throughout the history-match period, with those of a single well highlighted for comparison

Coming to the vector WTPCB_3, WTPCA_3 and WTPCT3 in Fig. - 10 for all the wells from start of simulation to about 2013, we see the contribution to C1N2 from reservoir oil (WTPCB_3; Below GOC) is 100% from a large share of the wells for majority of time.

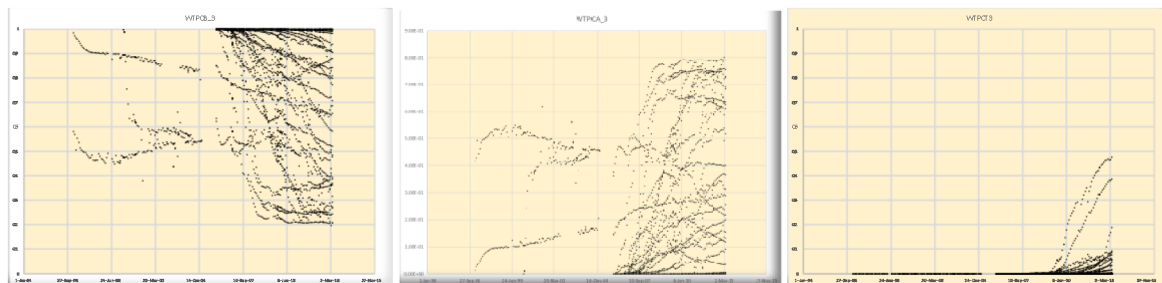


Figure 10—Production Concentrations of each of the sources (origins) to the component wzmf_3, and its evolution in time.

As simulation progress, contribution from Gas Cap (WTPCA_3; Above GOC) increases, reaching 80% for a few wells at later time. Contribution from Injection Gas (WTPCT3) is essentially 0% till about 2007. Till 2013 it never exceeds 50% for any well. Filtering out all wells except the example well, the contribution from each source vs. time is shown in Fig. - 11.



Figure 11—Well Tracer Production Concentrations for component wzmf_3 from the 3 sources (origins) for a single well.

The final plot is the sum of contributions (WTPC) (for component C1N2 only) from Above, Below and Inj. Gas which we expect to be 1.00 exactly, but the actual result appears close but not exact. We will discuss this under the section of numerical dispersion and its remedy.

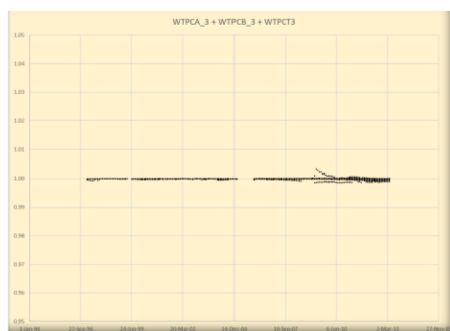


Figure 12—Analysis of sum of production concentrations for component wzmf_3 for the full HM period.

NUMERICAL DISPERSION

Since we are interested in using the latest results for each well (last producing time step) to the end of history match, which was 31-Dec-2023, the integration utility extracts data beyond 1-Jan-2022 only before importing into eGOR.

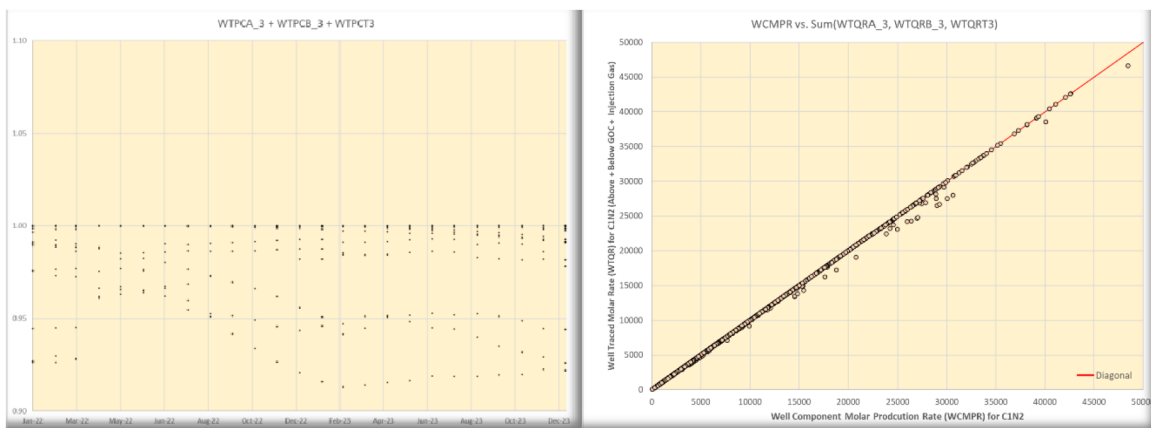


Figure 13—Numerical dispersion observed in values of Tracer Production Concentration and Tracer Molar Rates for the end of HM period

To overcome this issue, and recognizing that $WCMPR_n$ are the correct values, we apply a correction factor of:

$$\frac{WCMPR_n}{\sum(WTQRA_n + WTQRB_n + WTQRT_n)}$$

To the original WTPC values to compute "normalized" production concentration values for use in eGOR. Plotting these values and the corresponding molar rates results in the plots illustrated in Fig. - 14 where all the dispersion has disappeared.

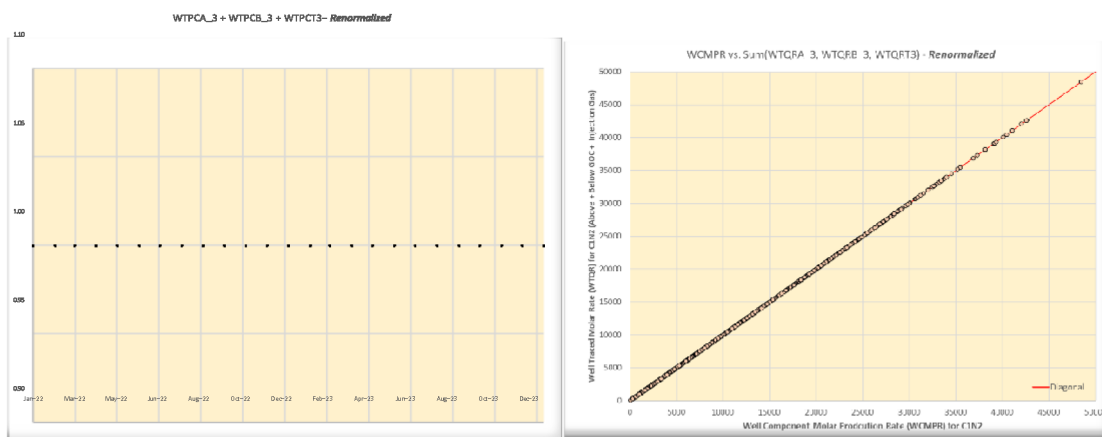


Figure 14—Effect of normalization to remove the observed numerical dispersion

UTILIZATION

A utility has been developed to make it simple to extract the above raw data from output of E300 to a csv file that will be read and imported by eGOR to populate the database. This utility is based on Python and runs on the OneHPC environment and utilizes the command run console instance of Aspentech/Emerson Tempest View for which a run time license is needed. Since the session is executed in the command line, the session lasts a few second to a minute at the most. For an explanation of these please see the section towards the beginning of this document

Once this utility is run, a csv is generated containing the information which is easily imported into Excel for visualization and review. For example, the WTPCB_3 for the first reported well is plotted in Fig. - 15.

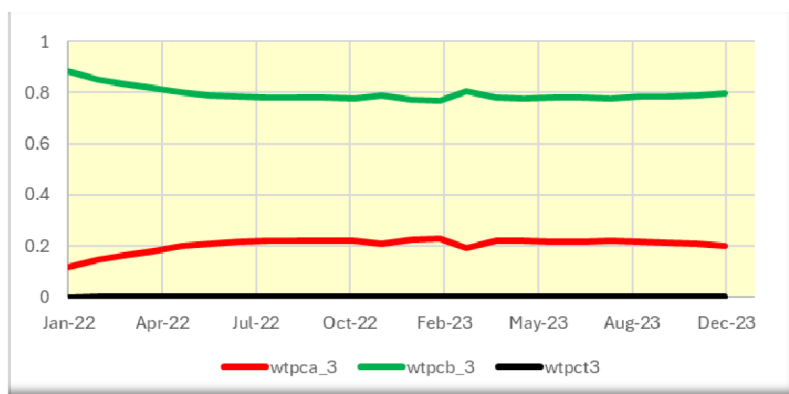


Figure 15—Production Concentration of example well plotted in Excel for visualization.

The last producing time step values will be used for eGOR which are:

wtpca_3	wtpcb_3	wtpct3
0.195712	0.80407	2.18E-04

This means, for component #3, which happens to be C1N2 for the example reservoir simulation run, 80.4% is sourced from below GOC (i.e. Reservoir Oil) 19.6% from above GOC (i.e. Gas Cap) and essentially nothing from Injection Gas. So, if the mole fraction of component #3 (WZMF_3) is 0.51926565, the following is the mole fraction of C1N2 from each respective source:

Above GOC	Below GOC	Inj. Gas
0.101626	0.417526	0.000113

In fact, this is the exact same calculation the eGOR import utility performs to compute the extended (e.g. EOS24) composition from the original (e.g. EOS8) composition for each well at end of history. This is detailed in a later section.

The period of 2 years is chosen, instead of the absolute last time step of the History Match (HM) run, to capture other wells which might have stopped production earlier.

Another detail worth noting is that production concentration of all components might not necessarily be the same. To illustrate this point, the plot in Fig. - 16 added the WTPC for component #5 (C4C5) to the same plot, while removing the series for injection gas (since the contribution is essentially zero anyway) and show the somewhat different production concentrations and changing with time.

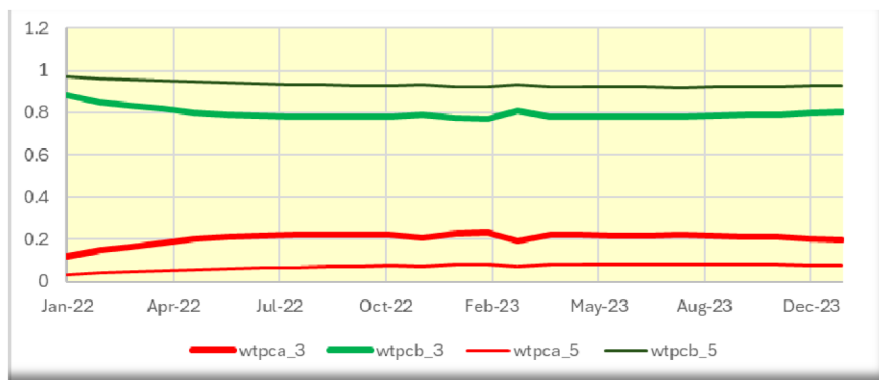


Figure 16—Well Tracer Production Concentration for 3rd & 5th components of the EOS to illustrate that these are not the same for all components.

DYNAMIC REFERENCE COMPOSITIONS

Once a format conversion is performed from E300 to a PRP file, eGOR has sufficient information to use the output from an E300 run, converted to csv. The basic information that is extracted (without delving into the complexity of source tracking for the moment) is the **wzmf** (well Total Mole Fraction) of each well for the last producing timestep. This provided the best basis of the representative reference composition for each well for use by eGOR, instead of the static, by-reservoir, compositions used in the original design. The by-reservoir compositions are from original PVT reports and do not represent the current composition of the well. The reservoir simulation integration for eGOR source tracking provides an opportunity to use these compositions in eGOR even if source tracking is not utilized.

As more reservoirs and fields incorporate integration between eGOR and reservoir simulation, more eGOR transformations will start using these "dynamic" reference compositions, since they come from dynamic simulation. As newer history matched runs are available, eGOR will benefit from the update reference composition, thus becoming truly dynamic.

Another aspect of dynamic composition is the ability to compute a reference composition for well which are new and not yet a part of the history match simulation. Fig. - 17 plots the composition of C1N2 as a function of GOR.

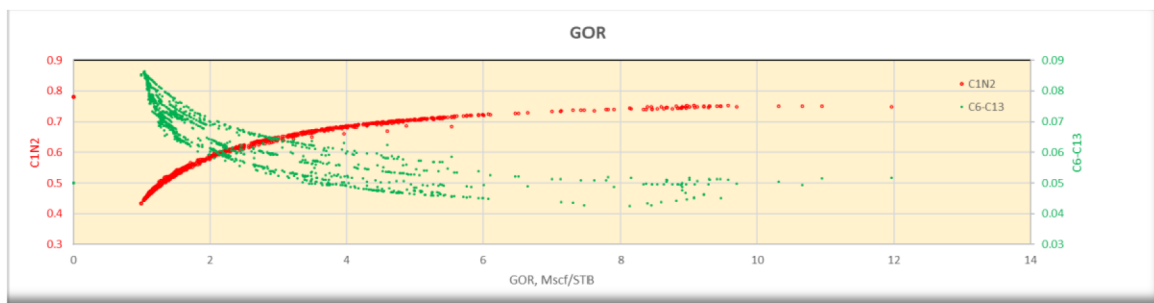


Figure 17—Relationship of GOR and mole fraction of wzmf_3

We see the general non-linear nature of the curves. This collates all the wells for all the time-steps on a single plot. However, if we choose to plot OGR (Oil Gas Ratio = 1/GOR) instead, this plot (Fig. - 18) suddenly takes a linear character.

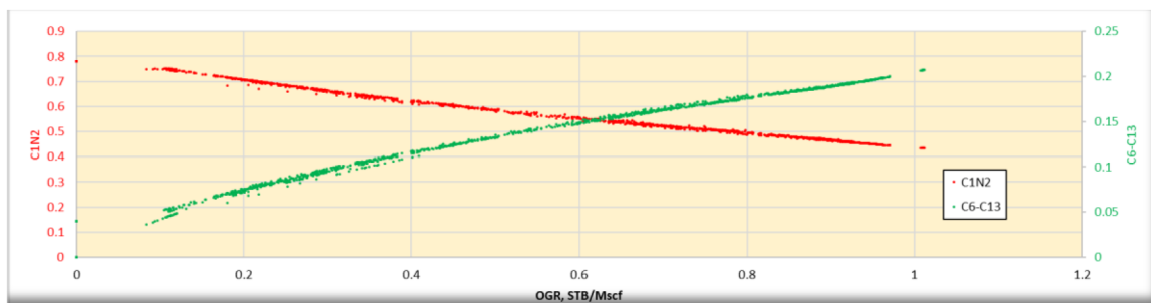


Figure 18—Linear Correlation of wzmf_3 mole fraction and OGR = 1/GOR.

eGOR populates a table with this data, albeit for the last production timestep on, which plots as follows:

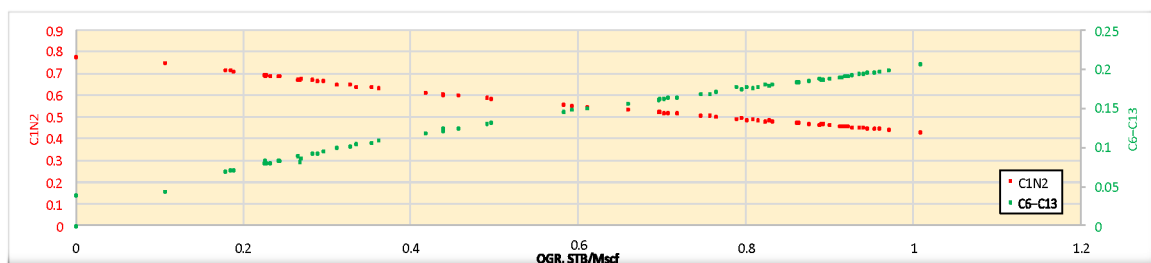


Figure 19—wzmf_3 & wzmf_6 vs. OGR correlation for the last production time step.

It has the same character and range of OGR. Due to this linear character, eGOR can easily use linear interpolation to estimate the reference composition of wells that were not included in the simulation, adding another reliable source of more representative reference composition for use by eGOR.

It should be noted that reference composition is the initial feed used by eGOR for Target GOR matching, which improves the representative (actual) flowing composition of the well at the time of well test.

EOS MODEL EXTENSION (EOS8 to EOS24) AND QC

For the purpose of EOS simulation with tagged components, a utility was developed to expand the PRP file from *nc* components to 3 *

nc component. All the properties of the tagged components are repeated except their name being prefixed with an "a-", "b-", or "i-".

Once imported into Resolve, or any other PetEx package, it resembles the snapshot in Fig. - 20, where 8 component EOS is "converted" to a 24-component one and imported:

Figure 20—"Extended" EOS model imported into PVTp

Similarly, the Binary Interaction Parameters triplicated.

A novel technical solution is used to merge the two key information into a single composite information that can still use the standard eGOR workflow (TargetGOR match followed by Field PTS transformation):

- Representative well reference composition (WZMF).
- Production Concentration of each component from the originating source (WTPCA, WTPCB, WTPCT)

An extended (24-component) reference composition is computed for each well from the above using the following formula:

$$wzmf_{24} = (wzmf_8 * wtpca) \sim (wzmf_8 * wtpcb) \sim (wzmf_8 * wtpct)$$

where the symbol $\sim$ indicates concatenation. This is illustrated in Fig. - 21 where original wzmf_3 are split into 3 tagged wzmf_3 components (e.g. A-C1N2, B-C1N2, and I-C1N2).

EOS8			EOS24		
CO2	wzmf_1	0.030	WTPCA_1	0.075	A-CO2 0.002248 =wzmf_1 * WTPCA_1
H2S	wzmf_2	0.000	WTPCA_2	0.130	A-H2S 5.9E-07 =wzmf_2 * WTPCA_2
C1N2	wzmf_3	0.000	WTPCA_3	0.077	A-C1N2 0.053104 =wzmf_3 * WTPCA_3
C2C3	wzmf_4	0.100	WTPCA_4	0.069	A-C2C3 0.007442 =wzmf_4 * WTPCA_4
C4C5	wzmf_5	0.055	WTPCA_5	0.048	A-C4C5 0.002654 =wzmf_5 * WTPCA_5
C6-C13	wzmf_6	0.083	WTPCA_6	0.040	A-C6-C13 0.003331 =wzmf_6 * WTPCA_6
C14-C29	wzmf_7	0.030	WTPCA_7	0.023	A-C14-C29 0.000703 =wzmf_7 * WTPCA_7
C30P	wzmf_8	0.006	WTPCA_8	0.013	A-C30P 8.51E-05 =wzmf_8 * WTPCA_8
			WTPCB_1	0.392	B-CO2 0.011757 =wzmf_1 * WTPCB_1
			WTPCB_2	0.870	B-H2S 3.94E-06 =wzmf_2 * WTPCB_2
			WTPCB_3	0.000	B-C1N2 0.206231 =wzmf_3 * WTPCB_3
			WTPCB_4	0.440	B-C2C3 0.047333 =wzmf_4 * WTPCB_4
			WTPCB_5	0.494	B-C4C5 0.027177 =wzmf_5 * WTPCB_5
			WTPCB_6	0.739	B-C6-C13 0.06139 =wzmf_6 * WTPCB_6
			WTPCB_7	0.977	B-C14-C29 0.029407 =wzmf_7 * WTPCB_7
			WTPCB_8	0.987	B-C30P 0.00627 =wzmf_8 * WTPCB_8
			WTPCT1	0.528	I-CO2 0.015834 =wzmf_1 * WTPCT1
			WTPCT2	0.000	I-H2S 0 =wzmf_2 * WTPCT2
			WTPCT3	0.613	I-C1N2 0.421748 =wzmf_3 * WTPCT3
			WTPCT4	0.488	I-C2C3 0.052519 =wzmf_4 * WTPCT4
			WTPCT5	0.457	I-C4C5 0.025137 =wzmf_5 * WTPCT5
			WTPCT6	0.220	I-C6-C13 0.018307 =wzmf_6 * WTPCT6
			WTPCT7	0.000	I-C14-C29 0 =wzmf_7 * WTPCT7
			WTPCT8	0.000	I-C30P 0 =wzmf_8 * WTPCT8

Figure 21—Generation of reference composition of the "extended" EOS model from output vectors of E300

So, any composition from E300 for a well can be converted from 8-component to 24-component. One QC that is necessary that the behavior of the two EOS models, with the associated reference compositions should be the same. To confirm, we perform a Phase Envelope calculation using both for two fluids:

- Original fluid composition from the end of history for an example well
- Adjusted fluid composition after Target GOR matching for the same well from a recent well test.

The results are identical, giving confidence in the conversion from EOS8 to EOS24:

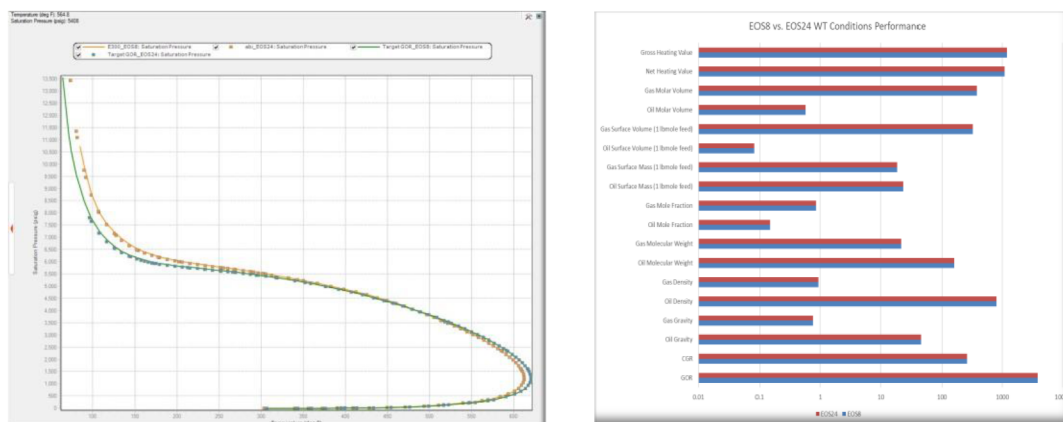


Figure 22—QC to confirm the phase envelope (hence the PVT behavior) of the original (8-component) EOS model and the "extended" (24-component) EOS model is the same. Two compositions are used, reference and Target GOR matched. Several other properties are computed with the two (original & extended) EOS models confirming the identical PVT behavior

Similarly, the property calculations for the well test for the two EOSes are also compared, since these results are more relevant than the Phase Envelope.

Virtual Flow Meter / Choke Model

Wells in the example field, which is the subject this work, are equipped with wellhead pressure and temperature (WHP, WHT), choke valve readings, flowline pressure and temperature (FLP, FLT). Each well in the cluster is connected to dedicated MPFM. All the data from these sensors and the real time data from MPFMs in all clusters are available in FDIS. This field qualifies as a digital or instrumented field and provides an opportunity to implement further technical solutions.

One of the methods to estimate oil and gas rates was proposed by Mohammad Ali Kargarpour in his paper: "Oil and gas well rate estimation by choke formula: semi-analytical approach" in 2018.

With reference to Fig. - 23, the equations are show in Fig. - 24.

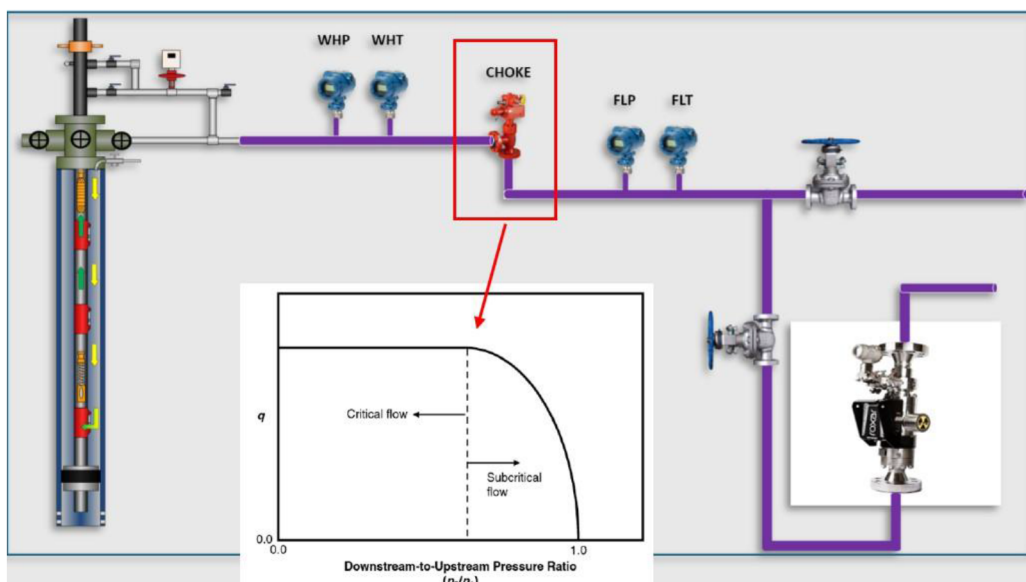


Figure 23—Typical Well Head configuration of an oil well and the parameters used in the Choke Formula

Critical flow → $\text{if } \frac{FLP}{WHP} \leq 0.55:$

$$Q_{Liq} = CF \times WHP \times Choke_{inch}^2 \times \left[\frac{\sqrt{WHP}}{552 \times \sqrt{\frac{(1-FLP/WHP)}{SG_{Liq}}}}} + \frac{GLR}{14387} \right]^{-1}$$

Subcritical flow → $\text{if } \frac{FLP}{WHP} > 0.55:$

$$Q_{Liq} = CF \times WHP \times Choke_{inch}^2 \times \left[\frac{\sqrt{WHP}}{552 \times \sqrt{\frac{(1-FLP/WHP)}{SG_{Liq}}}}} + \frac{GLR}{65554 \times \sqrt{\left(\frac{FLP}{WHP}\right)^{1.5625} \times \left(1 - \left(\frac{FLP}{WHP}\right)^{0.21875}\right)}}} \right]^{-1}$$

Figure 24—Analytical expression to compute the liquid rate for Critical & Subcritical flow conditions

The critical and subcritical flow regimes are separated by FLP/WHP ratio equal to 0.55, i.e. $\frac{FLP}{WHP} = 0.55$

In the above equations, CF stands for Correction Factor or Tuning Factor. This parameter is used to tune the model to MPFM measurements during a well test. Note that this linear tuning method is not perfect. It is well known that the processes in wellbore and across choke valve are complex and thus non-linear. However, this simple linear tuning of the model is a good starting point.

The above method was developed based on test data. The range of variables in this implementation is as follows:

Table 2—Range of variables in which the Choke formula is applicable

API	12.8 – 42
WHP	38–5538 (psi)
FLP/WHP	0.01–0.91
GOR	61–6044 (scf/stb)
Choke size	8/64–96/64 (inches)

Fig. - 25 illustrates the proposed tuning process for all the wells in this field:

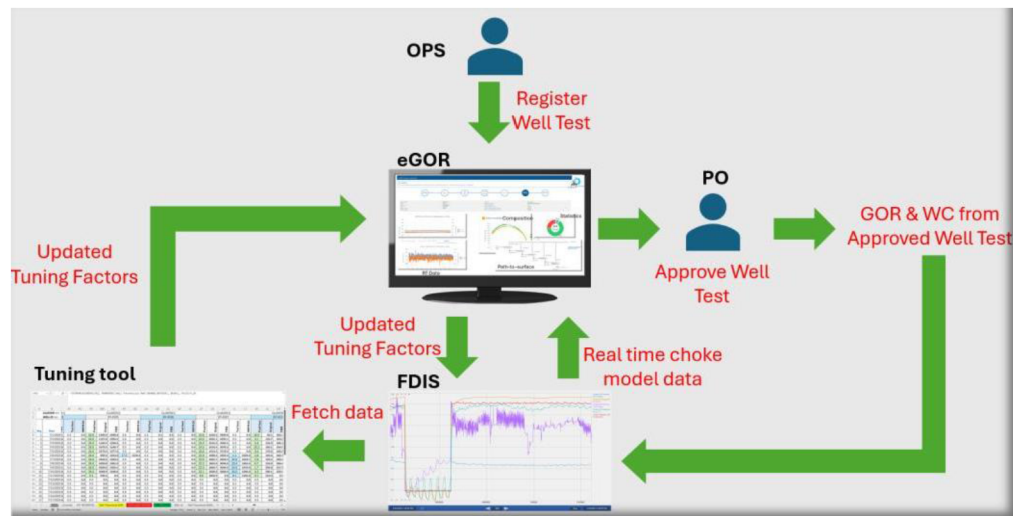


Figure 25—Proposed tuning workflow to ensure the VFM (choke models) are tuned to latest approved well tests

Limitations:

- The model does not consider the line mixture density variations.
- Linear correction factor might be not optimum correlation factor
- The critical flow ratio (FLP/WHP) is taken as constant value = 0.55. However, the actual ratio depends on fluid properties, upstream and downstream conditions.
- The choke model/method used in this POC is designed for certain data set which might not be always representative of real behavior in the field
- Since GOR and WC are considered as constants in the method, these parameters need to be updated with time.
- Dependency on field sensors data quality:
 - Stuck choke values
 - WHP, FLP and choke malfunctioning
 - FDIS availability
- New wells are not mapped to FDIS:

Benefits:

- Increased utilization of real time data from sensors
- Continuous feedback system for well testing validation process
- Well testing anomaly detected on time

Results:

Even with simple tuning process, the results are sufficiently good to continue further improvement to the methodology.

The tuned choke models were used to re-calculate the field total theoretical rates (sum of wells) and compare against field actuals and the improvement over the Legacy methods are significant.

Real Time (Online) VFM Implementation

AVEVA PI System is an industry standard system to capture real time data generated by measurement systems implemented in the fields (e.g. pressure, temperature, flow transmitters) and make it available to a wider audience in the company, fully compliant with the access protocols. ADNOC Onshore has rebranded this system as **FDIS** which is the common terminology within the company to all systems related to real time data.

In ADNOC Onshore, all fields are categorized into two based on the degree of instrumentation employed in the respective production systems:

- a. Instrumented Fields are those that have a minimum set of parameters being monitored by means of instrumentation on each well. These include the Well Head Pressure (WHP), Well Head Temperature (WHT), Production Choke Opening etc. In addition to the well specific parameters for all the wells, all well testing stations (e.g. Multi Phase Flow Meters – MPFM) are also being captured and available in FDIS.
- b. Non or Partially Instrumented Fields are those where only a few, or none, of the wells have online transmitters for WHP, WHT etc. However, the well testing measurement stations are usually 100% instrumented and the real time flow rates and associated line conditions (pressures and temperatures) are available in FDIS.

FDIS ecosystem supports several client/user level tools that provide different functionality and solutions for the end user.

- PI DataLink is an Excel Addin that allows directly querying of FDIS Tags for monitoring, QC and system health purposes. eGOR support team uses this extensively to verify proper functioning of the Tags identified for each parameter of each well/meter of each field. This is an invaluable tool to support robustness & reliability of the FDIS system for eGOR's input from FDIS real time data.
- PI Vision: Using these visualizations provide Field specific monitoring dashboards to provide at a glance status of the health of these Tags, specifically to identify periods of communication loss.

Apart from getting data from the on-field instrumentation, via the OPC servers, FDIS also provides a means of creating Tags that are categorized as Calculations Tags (also called Performance Equation). These Calculation Tags can have elaborate equations based on one or several other tags

With the collaboration of FDIS Team, a successful attempt was made to use FDIS to configure the Choke Model equation as described in the previous section. This section describes how it was done and its building pieces.

- A. Real Time FDIS Tags: The choke model equation used the following real time tags for each well as input to the equation:
 - a. Well Head Pressure (WHP)
 - b. Flow Line Pressure (FLP)
 - c. Choke Opening (Choke)
- B. VFM_tuning_factor FDIS Tags: The values for these Tags for each well is determined in an offline tuning exercise and uploaded to FDIS on adhoc basis when these are observed to change.
- C. Well Test basis FDIS Tags: These Tags are automatically update by eGOR after a well test has been approved in eGOR.
- D. eGOR Transformation FDIS Tags: These Tags are also updated by eGOR after the well test has been approved in eGOR. These are numbers that relate the Vendor (MPFM rates) to the eGOR Transformed (Corrected) Rates and the associated GOR.

- E. **Choke Model (VFM) Equation FDIS Tags:** The Oil Theoretical Rate is implementation of the Choke equation. This is the primary Tag which is used to update secondary tags of Gas & Water Theoretical Rates and the "Transformed" Rates.

Each of the above Tags are prefixed with Well Name/Number. There was heavy usage of Excel and its PI_DataLink to generate and verify the Choke Model expression before it was shared with the FDIS Team for configuration of individual FDIS Tags. Several iterations of back and forth, with corrections, were needed to finalize the expressions.

PI Vision dashboards were established for collection of tags for individual wells. There are 18 Tags for each well for the "normal" solution and 24 Tags for the "source tracking" solution, which we discuss in the next section.

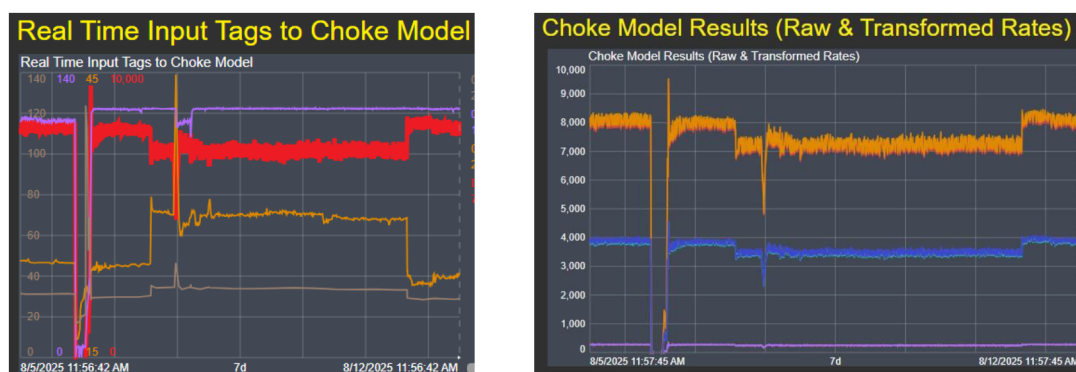


Figure 26—Real Time Tags and resulting Oil & Gas Theoretical Rates computed by FDIS expression for example well

Daily and Reservoir/Field Level Aggregation

The VFM Solution of Example Reservoir A covers all the wells of all reservoirs of the field. The raw Tags provide real time values based on real time values of well head parameters and the currently effective values of other parameters from the last approved Well Test and eGOR transformation.

To obtain Daily values from these real time Tags, a daily aggregation routine is setup in eGOR which runs on a schedule. It generates daily averages based on time-weighted averaging functionality of FDIS. The results are stored in an eGOR database and exposed to a Spotfire visualization where daily and monthly rates are automatically available on well and reservoir/field levels.

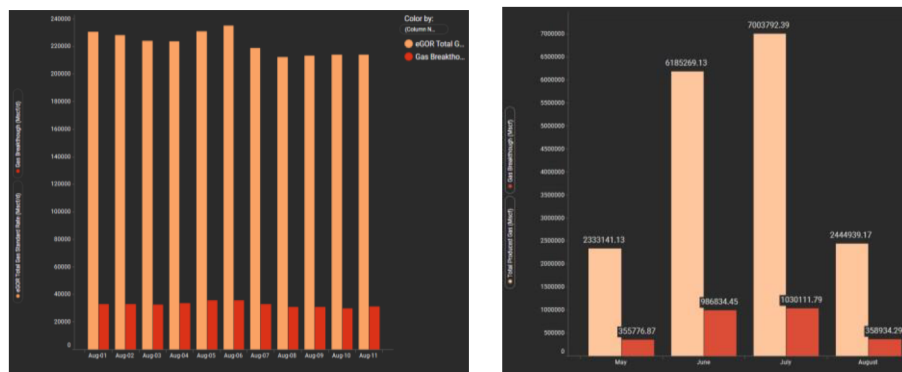


Figure 27—Daily and Monthly aggregated Total Gas Rates alongwith Inj. Gas Breakthrough rates

Combining VFM with Source Tracking

The VFM Solution of Example Reservoir A covers all the wells of all reservoirs of the field. With the implementation of source tracking in Example Reservoir A, we were able to extend this daily well/reservoir results to compute oil and gas source segregated results.

To implement this, we created an additional 6 FDIS Tags for the 64 wells belonging to Example Reservoir A:

Tag	Quantity
_Gas_Source_A	Fraction of Surface Gas from Source Above GOC
_Gas_Source_B	Fraction of Surface Gas from Source Below GOC
_Gas_Source_I	Fraction of Surface Gas from Source Injection Gas
_Oil_Source_A	Fraction of Surface Oil from Source Above GOC
_Oil_Source_B	Fraction of Surface Oil from Source Below GOC
_Oil_Source_I	Fraction of Surface Oil from Source Injection Gas

These fractions are computed for each well once the well is processed by eGOR using Source Tracking Option, and the results are approved by the relevant authority, they automatically populate the Tag, effective from the date/time of approval of the well test.

Further, the following 2 Tags are updated using the real time values of Transformed Gas & Oil rates:

_ChokeModel_BreakthroughGas	'_ChokeModelTransformed_Gas' * '_Gas_Source_I'
_ChokeModel_Condensate	_ChokeModelTransformed_Oil' * '_Oil_Source_A'

Although these Tags give real time values of the Breakthrough Gas (Gas Injection source) and Condensate (Gas Cap source), daily values are more usable and presentable. For this purpose, these calculations are implemented in Spotfire, which accesses the daily averaged data stored on eGOR database. Daily Rates for all three sources are segregated and presented for each well in the following dashboards:

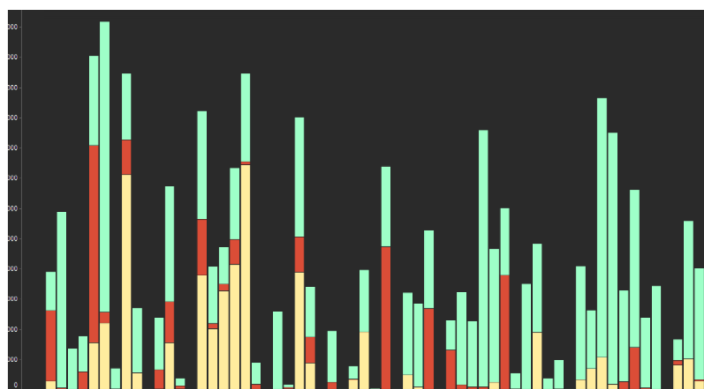


Figure 28—Segregated Volumetric Gas Rates for each well for any particular day of the month - automatically generated from RT VFM Tags

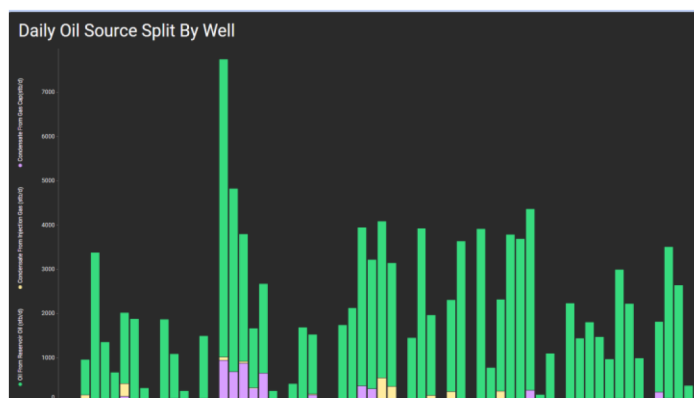


Figure 29—Segregated Volumetric Oil (Condensate) Rates for each well for any particular day of the month - automatically generated from RT VFM Tags

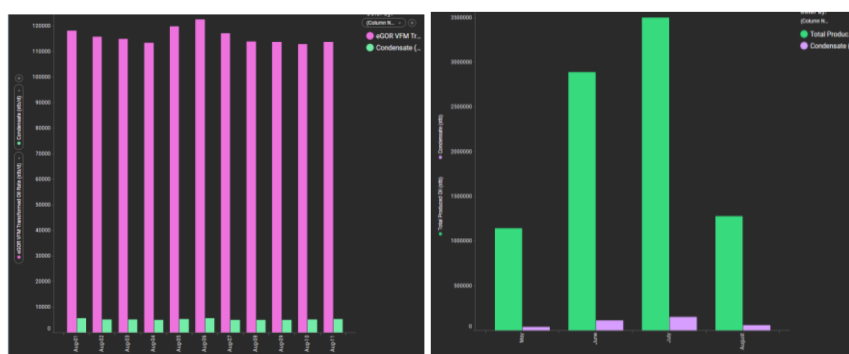


Figure 30—Daily and Monthly Total Oil and Condensate (from Gas Cap) generated using RT VFM Tags

Conclusions

Through integration of the following four distinct engineering applications and tools:

- **Reservoir simulation** to quantify contribution of each source to the molar production of each component in each well
- **EOS based Surface Separation simulation** to track the molecular contribution of each source to each surface product while matching the GOR recorded at every well test.
- **Virtual Flow Metering** (Choke Model) using published and industry embraced method of estimating Oil/Liquid rates from real time well head parameters
- **Real Time parameters capture & visualization system** (Aveva PI) to adapt it for real time VFM calculations and aggregation on daily well, reservoir and field levels

we have demonstrated the deployment of a system that provides:

- **Daily estimates of well oil, gas & water rates** on days other than when the well was tested on. This provides an important and missing information: well rates on other days were forced to be at same level as on the well test day. This indirectly provides more representative –daily rates for use in well allocation resulting in better reconciliation factors.
- **Daily estimates of source segregation and quantification** of the daily rates from three sources. Aggregating on monthly level and converting rates to volumes will provide the Injection Gas Breakthrough amounts as a reliable basis for fiscal implications. Further, estimations of daily/monthly production of condensate from Gas Cap origin also have enormous fiscal implications.

Way Forward

To implement reservoir simulation integration for all reservoirs of all fields. Also implement VFM solution for all wells of all fields as the fields embrace digitalization and become fully instrumented.

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